

For advancing functional genomics with a wide range of applications on cell engineering, inferring gene regulatory networks, and also in general revealing gene functions. So, in particular, in perturbation response prediction tasks, we have a group of, typically, we have a group of, so where was my mouse? Okay. So, we have a group of control cells, then we have information about the perturbation that is being applied, and a goal is to predict the post-perturbation profile. So, we want to answer a question, what would happen if we were to perturb a single cell in a certain way? Perturbations can be genetic and be chemical perturbations, but in this talk, I will focus mainly on the, I will completely focus on genetic perturbations. Meaning, we want to predict if a certain gene has been perturbed, what will be the effect of this perturbation. Now, in perturbation response prediction task, you can imagine that you have some trained perturbations, you have also a group of control cells, and now you want to predict how cells will respond to these test perturbations. And the recent challenge for advancing functional genomics with a wide range of applications on cell engineering, inferring gene regulatory networks, and also in general revealing gene functions. So, in particular, in perturbation response prediction task, we have a group of, typically, we have a group of, so where was my mouse? Okay. So, we have a group of control cells, then we have information about the perturbation that is being applied, and a goal is to predict the post-perturbation profile. So, we want to answer a question, what would happen if we were to perturb a single cell in a certain way? Perturbations can be genetic and be chemical perturbations, but in this talk, I will focus mainly on the, I will completely focus on genetic perturbations, meaning we want to predict if a certain gene has been perturbed, what will be effect of this perturbation. Non-perturbation response prediction task, you can imagine that you have some trained perturbations, you have also a group of control cells, and now you want to predict how cells will respond to these test perturbations. And recently, so is it possible, because I don't think, no, my end of, okay. So, okay, there is a box here which says that there have been recently methods that are aimed to kind of predict how cells will respond to unseen genetic perturbations. In order, this means that this kind of genetic perturbation, we want to predict the response for some genes that we have not seen, the genes have not been perturbed in this trained data set. And in this case, of course, we need some prior knowledge in order to predict how cells respond to these unseen genetic perturbations. So, there have been kind of two groups of methods. One groups of methods are these big foundation models, where this prior comes from pre-training or large-scale single-cell RNA sequencing data sets, and more specialized models that kind of directly incorporate some priors, for example, functional similarity of genes. So, in our team, recently, in collaboration with Sarat Heinrichs and Morniten lab, and this has been led by my postdoc, Ramon, we asked the question, to what extent can we predict transcriptional outcomes of unseen genetic perturbations? So, we designed a 4-0 benchmarking study where we took 10 data sets across three different technologies, including CRISPR-I, CRISPR-CoI, and CRISPR-A, and five different cell lines. And we used the standard metrics that had been widely adopted in a field for perturbation

response prediction tasks, in particular, Pearson correlation between predicted and ground truth expression changes of a perturbation with respect to a control cells. So you have basically control cells, and then you predict the post-perturbation profile, you subtract the control cells, see the differences, and you compare to this with the ground truth that you have. And you can compute this for all genes, so you can, for example, focus on a subset of differentially expressed genes. And we constructed a very simple baseline that we call a perturbed mean, where basically, as your prediction, you take the average expressions of all perturbed cells as a prediction. And in this case, you use this profile as a prediction for all unseen genetic perturbations. So, this means that all non-seen genetic perturbations receive exactly the same prediction, which corresponds to this perturbed mean, which is this average expression of all perturbed cells. So, surprisingly, we find that this very simple baseline is competitive to state-of-the-art methods. So, it often outperforms these methods or achieves very close performance across these 10 different data sets and also across different metrics. So, this is all one gene perturbations. We were wondering, is this the case also for combinatorial perturbations? So, for combinatorial perturbations, we could design another also pretty simple baseline where we just incorporate additional priors. For example, if I'm predicting, if I haven't seen any of these genes being perturbed, then I would just take this average expression, so it would be exactly the same as the baseline I introduced before. If I have seen one of these genes, for example, being perturbed in my trained data set, I can do something like smarter, but still, I just take the average expression of cells in my trained data set and take these two averages of this perturbed mean that I introduced earlier, and centroid of these cells that underwent the same type of perturbation. And if I have seen both of these perturbations, genetic perturbations, then I can just take their average. And we show actually that this baseline, so it's shown here in up. So, it's shown here in brown color. So brown, so the legend is that the brown color is this matching mean, and the purple color is this perturbed mean that I introduced before, and all other methods, so other colors correspond to different baselines, CPA, gears, GPT, and fine-tuned version of the GPT. And we see actually that this simple baseline, this data introduced matching mean, significantly outperforms on this widely used Norman data set that has combinatorial perturbations. So, we were wondering why this happened, you know, why this kind of these simple baselines perform so well, and what is the reason behind this. So, we hypothesize that the reason may be in systematic variation or kind of systematic differences between the control and perturbed cells. These systematic differences may be caused, for example, by potential selection biases, that we are kind of perturbing sets of genes that are functionally related, that is kind of causing systematic differences between the control and perturbed cells, may be also caused, for example, by underlying biological factors such as cell cycle. And I will show example for one data set, but we did like investigations for different data sets that I introduced earlier. So, for example, on this kind of genome-wide perturbation data set, we can see actually that cells, that if you look at the distributions of cells across different phase density, across different

cell cycle phases, we can see that there are significant differences. For example, in the control cells, there are about 25% of cells in the G1 phase, while in the perturbed cells, there are about 46% of cells in the G1 phase. And this can be actually explained in this data set through chromosomal instabilities that are induced by perturbations, and RPI1 cells that are p53 positive correspond to this through cell cycle arrest. So, we next aim to quantify this degree of systematic variation. So, in particular, we define average perturbation shift as a vector that points from the control cells to the centroid of perturbed cells. And we define this perturbation-specific shift as a vector that points from the control cells to the group of cells that underwent the same type of perturbation. Now, we can measure the cause and similarity, and we can measure a look at the causal similarity between these different perturbation shifts to kind of quantify these systematic differences. And if all these vectors point in a similar direction, this would kind of like be mean that the vectors are aligned kind of in similar directions, meaning kind of there is a, they are all very similar to each other and kind of points to possibly non-specific shifts in the gene expressions that are rather caused by systematic differences. So, if you look now, measure the systematic variation across different data sets, what we see actually that Adams on the Norman data set that are very frequently used to benchmark show very high degree of systematic variation. And interestingly, if we measure now on the x-axis, we show different data sets, and here data sets are ordered by the amount of systematic variation show on top. And now, if I measure Pearson correlation coefficients or kind of performance of different methods with respect to amount of the systematic variation, you can see there is a very high correlation, meaning kind of higher performance is typically caused by a higher degree of systematic variation across these different methods. So, what this points us to is that commonly used metrics can be influenced by this systematic variation and also may lead to inflated performance. So, in order to do this, in order kind of motivated by this, we propose the systema evaluation framework where we redefine the evaluation metrics to rather focus on perturbation-specific effects rather than this kind of systematic variation, to control for this systematic variation. So, before when we compute the Pearson correlation, one would take control cells as a reference. What we propose to do in Systema is take the perturbed centroid as a reference, and now I would focus on shifts from perturbed reference in order to capture the perturbation-specific effects. And now, if we evaluate again methods using this new metric with the change reference, where now I'm using perturbed centroid as a reference, whether the centroid of control cells, we can see that perturbed mean achieves almost zero performance, which is actually expected. But we can also see that now the performance is also significantly degraded by existing methods. So, what this points us to is actually that predicting the perturbation response to unseen genetic perturbation is a very difficult task. Yeah, that sounds good. So, now if we also look at the, again, if you do the same exercise and now measure, quantify the systematic variation and measure now this new Pearson correlation with respect to the perturbed centroid as a reference, now we can see that now it's a local, we have a low correlation with

the degree of systematic variation across all these different data sets. In addition, in this system evaluation framework, we also introduce a metric which is kind of quite intuitive and simple. It's called centroid accuracy. So, it tells us for a given perturbation, is the predicted centroid closer to the actual centroid or closer to other centroids? And what this means is basically that centroid accuracy of one indicates that the profile recovers the expected transcriptional effects of the perturbation, because my prediction would be the closest to the ground true, closer to the ground truth centroid compared to all other centroids that I can measure. Now, if we evaluate the centroid accuracy across different methods, we see again that, and here perturbed mean is shown as this dashed black line, we can see that perturbation scores, average perturbation scores barely exceed those of the perturbed mean, but out of existing methods, fine-tune version of single-cell GPT performs the best. Now, using the centroid accuracy, we can take desire, like method of our choice, and then we can compare the centroid accuracy of this method versus centroid accuracy of the perturbed mean. And now I can divide the predictions in perturbations in those that are informative, have informative predictions, meaning that I'm predicting something beyond the systematic variation versus uninformative predictions where I'm actually capturing something that I would already capture with this perturbed mean. If you look at this now metric for, for example, fine-tuned version of the single-cell GPT and what are the genes for which the model makes informative predictions, so these are mainly highly conserved ribosomal protein genes. Now, we can also use systema for kind of uncovering, for example, coarse-grain perturbation effects. So, we can define, this is an example of defining class-specific centroid, but you can define also other class-specific centroid. For example, here we define class-specific centroid based on those who have perturbations who have high chromosomal instabilities and those that have low chromosomal instabilities. And now we can actually measure whether my predicted centroid is to which of these two class-specific centroid my prediction is closer to. And now we did this evaluation across different methods, and we see actually that even in this kind of core, like distinguishing, learning this kind of coarse-grain perturbation effects, most of the method achieves near random performance, with the exception of a fine-tuned version of the single-cell GPT, which achieves about 0.7 AUC score. So, the paper is coming out soon, but the code is already online, and I think the paper should also be online very soon. And finally, I would like to acknowledge the work done by my, led by my post-Dr. Ramon, and in collaboration with Sarah and more, and also other students that contributed this, Macek, who is also here in the audience, if you have additional questions, and Zoe and Liangce and Sean. So, thank you so much. Is it interesting to try and assess it to calculate the ability of the prediction? The pseudo-belt doesn't matter because it's not the data, but you can try to capture the data. You mean kind of more kind of try to match cells beforehand or something like that? Let's say the most extreme case, let's say a prediction has a bi-model effect, uncertainty, just a priority. In some cells, it makes it very high, in some cells, it makes it very low. So, the pseudo-belt analysis, your model should always predict that it's

in some non-existent real state. Is that a realistic concern, or is the true answer? Yeah, it is. But, I mean, all of these metrics, of course, are based on kind of distributions, right? Because you will take a distribution of control cells, and then you take the distribution also of cells that have been kind of perturbed. But ideally, of course, you could maybe try to do some like there have been methods also kind of try to match cells, but then it's kind of matching is also quite inaccurate. So, of course, there is this problem that we are kind of evaluating things still on the population, on the populations of the cells. So, we have to move on, sorry, but we can ask later or again on the website. Thanks again, Mario. Thank you. Okay, so is it possible? Because I don't think, no, my end of, okay. So, okay, there is a box here which says that there have been recently methods that are aimed to kind of predict how cells will respond to unseen genetic perturbations. In order, this means that this kind of genetic perturbations, we want to predict the response for some genes that we have not seen, the genes have not been perturbed in this trained data set. And in this case, of course, we need some prior knowledge in order to predict how cells respond to these unseen genetic perturbations. So, there have been kind of two groups of methods. One group of methods are these big foundation models, where this prior comes from pre-training or large-scale single-cell RNA sequencing data sets, and more specialized models that kind of directly incorporate some priors, for example, functional similarity of genes. So, in our team, recently, in collaboration with Sarah Techmann's and Mornitsen's lab, and this has been led by my postdoc, Ramon, we asked the question, to what extent can we predict transcriptional outcomes of unseen genetic perturbations? So, we designed a 4-0 benchmarking study where we took 10 data sets across three different technologies, including CRISPR-I, CRISPR-Co, and CRISPR-A, and five different cell lines. And we used the standard metrics that had been widely adopted in a field for perturbation response prediction tasks, in particular, Pearson correlation between predicted and ground truth expression changes of a perturbation with respect to a control cells. So you have basically control cells, and then you predict the post-perturbation profile, you subtract the control cells, see the differences, and you compare to this with the ground truth that you have. And you can compute this for all genes, so you can, for example, focus on a subset of differentially expressed genes. And we constructed a very simple baseline that we call a perturbed mean, where basically as your prediction, you take the average expressions of all perturbed cells as a prediction. And in this case, you use this profile as a prediction for all unseen genetic perturbations. So, this means that all non-seen genetic perturbations receive exactly the same prediction, which corresponds to this perturbed mean, which is this average expression of all perturbed cells. So, surprisingly, we find that this very simple baseline is competitive to state-of-the-art methods. So, it often outperforms these methods or achieves very close performance across these 10 different data sets and also across different metrics. So, this is all one gene perturbations. We were wondering, is this the case also for combinatorial perturbations? So, for combinatorial perturbations, we can design another also pretty simple baseline where we just incorporate additional priors. For

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